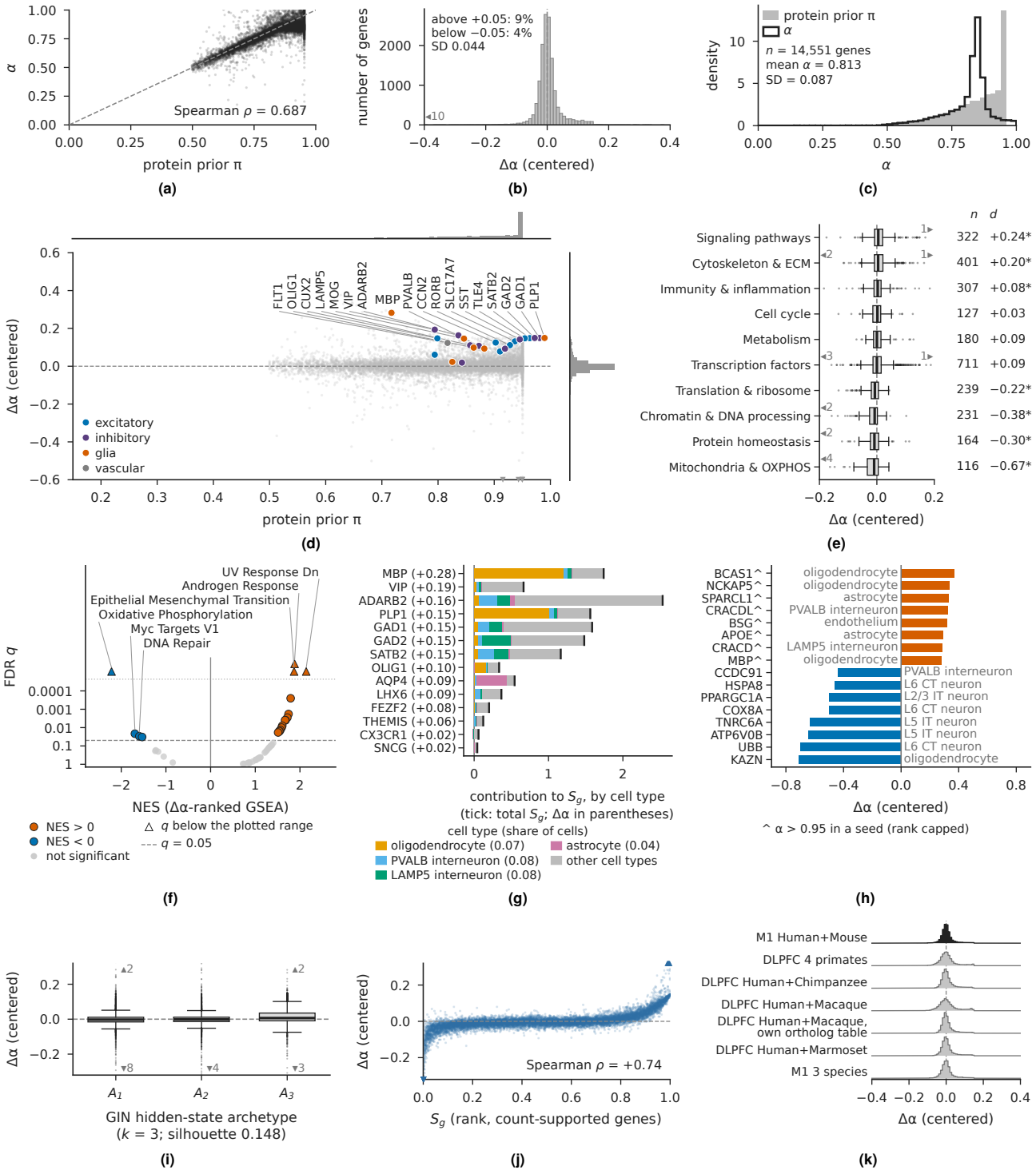

Spearman ρ = 0.687
protein prior π
α
(a)
above +0.05: 9%
below −0.05: 4%
SD 0.044
number of genes
Δα (centered)
(b)
protein prior π
α
n = 14,551 genes
mean α = 0.813
SD = 0.087
density
(c)
FLT1
OLIG1
CUX2
LAMP5
MOG
VIP
ADARB2
MBP
PVALB
CCN2
RORB
SLC17A7
SST
TLE4
SATB2
GAD2
GAD1
PLP1
excitatory
inhibitory
glia
vascular
Δα (centered)
protein prior π
(d)
n
d
Signaling pathways 322 +0.24*
Cytoskeleton & ECM 401 +0.20*
Immunity & inflammation 307 +0.08*
Cell cycle 127 +0.03
Metabolism 180 +0.09
Transcription factors 711 +0.09
Translation & ribosome 239 −0.22*
Chromatin & DNA processing 231 −0.38*
Protein homeostasis 164 −0.30*
Mitochondria & OXPHOS 116 −0.67*
Δα (centered)
(e)
UV Response Dn
Androgen Response
Epithelial Mesenchymal Transition
Oxidative Phosphorylation
Myc Targets V1
DNA Repair
FDR q
NES (Δα-ranked GSEA)
NES > 0
NES < 0
not significant
△ q below the plotted range
q = 0.05
(f)
MBP (+0.28)
VIP (+0.19)
ADARB2 (+0.16)
PLP1 (+0.15)
GAD1 (+0.15)
GAD2 (+0.15)
SATB2 (+0.15)
OLIG1 (+0.10)
AQP4 (+0.09)
LHX6 (+0.09)
FEZF2 (+0.08)
THEMIS (+0.06)
CX3CR1 (+0.02)
SNCG (+0.02)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
oligodendrocyte (0.07)
astrocyte (0.04)
PVALB interneuron (0.08)
other cell types
LAMP5 interneuron (0.08)
(g)
BCAS1^ oligodendrocyte
NCKAP5^ oligodendrocyte
SPARCL1^ astrocyte
CRACDL^ PVALB interneuron
BSG^ endothelium
APOE^ astrocyte
CRACD^ LAMP5 interneuron
MBP^ oligodendrocyte
CCDC91 PVALB interneuron
HSPA8 L6 CT neuron
PPARGC1A L2/3 IT neuron
COX8A L6 CT neuron
TNRC6A L5 IT neuron
ATP6V0B L5 IT neuron
UBB L6 CT neuron
KAZN oligodendrocyte
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.148)
A_1
A_2
A_3
(i)
Spearman ρ = +0.74
S_g (rank, count-supported genes)
Δα (centered)
(j)
M1 Human+Mouse
DLPFC 4 primates
DLPFC Human+Chimpanzee
DLPFC Human+Macaque
DLPFC Human+Macaque, own ortholog table
DLPFC Human+Marmoset
M1 3 species
Δα (centered)
(k)